

Financial statement analysis

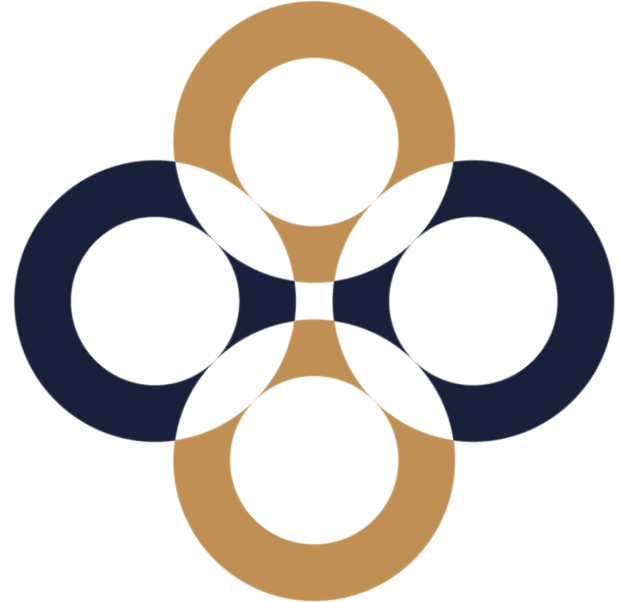
International Corporate Finance – 1

Kata Váradi, Leyla Yusifzada, Noémie Cabau

Corvinus University of Budapest, Institute of Finance

Based on:

Berk – DeMarzo: Corporate Finance. 6th edition, Pearson – 2nd chapter



Firms' Disclosure of Financial Statements

Financial statements:

accounting reports with past performance information that a firm issues periodically.

→ *source of information for: manager, investors, financial analyst, outside parties (eg. creditors)*

USA: file the statements with the Securities and Exchange Commission (SEC): **10-Q** (*quarterly*) & **10-K** (*annual*) + **annual report** to the shareholders

Common set of rules and standards: + auditor (*neutral third party*)

- Generally Accepted Accounting Principles (**GAAP**)
- International Financial Reporting Standards (**IFRS**)

Types of Financial Statements

1. Balance sheet (BS)
2. Income statement (IS)
3. Statement of cash flows (CF)
4. Statement of stockholders' equity

Balance Sheet I.

A snapshot in time of the firm's financial position.

$$\text{Assets} = \text{Liabilities} + \text{Stockholders' Equity}$$



What the company owns = what the company owes + (Assets – Liabilities)

Assets I.

Current Assets: Cash or expected to be turned into cash in the next year

- Cash

- Marketable Securities

- Accounts Receivable

- Inventories

- Other Current Assets

 - Example: Pre-paid expenses

Assets II.

Long-Term Assets

Net Property, Plant, and Equipment

Depreciation (and Accumulated Depreciation)

Goodwill and intangible assets

Amortization

Other long-term assets

Example: Investments in Long-Term Securities

Liabilities I.

Current Liabilities: Due to be paid within the next year

- Accounts Payable

- Short-Term Debt/Notes Payable

- Current Maturities of Long-Term Debt

- Other Current Liabilities

 - Taxes Payable

 - Wages Payable

Liabilities II.

Long-Term Liabilities

Long-Term Debt

Capital Leases

Deferred Taxes

Balance Sheet II.

GLOBAL CONGLOMERATE CORPORATION

Consolidated Balance Sheet Year Ended December 31 (in \$ million)

Assets	2022	2021	Liabilities and Stockholders' Equity	2022	2021
Current Assets			Current Liabilities		
Cash	21.2	19.5	Accounts payable	29.2	24.5
Accounts receivable	18.5	13.2	Notes payable/short-term debt	3.5	3.2
Inventories	15.3	14.3	Current maturities of long-term debt	13.3	12.3
Other current assets	2.0	1.0	Other current liabilities	2.0	4.0
Total current assets	57.0	48.0	Total current liabilities	48.0	44.0
Long-Term Assets			Long-Term Liabilities		
Land	22.2	20.7	Long-term debt	99.9	76.3
Buildings	36.5	30.5	Lease obligations	—	—
Equipment	39.7	33.2	Total debt	99.9	76.3
Less accumulated depreciation	(18.7)	(17.5)	Deferred taxes	7.6	7.4
Net property, plant, and equipment	79.7	66.9	Other long-term liabilities	—	—
Goodwill and intangible assets	20.0	20.0	Total long-term liabilities	107.5	83.7
Other long-term assets	21.0	14.0	Total Liabilities	155.5	127.7
Total long-term assets	120.7	100.9	Stockholders' Equity	22.2	21.2
Total Assets	177.7	148.9	Total Liabilities and Stockholders' Equity	177.7	148.9

Balance Sheet III.

1. Net Working Capital = Current Assets – Current Liabilities
= Cash + Inventory + Receivables – Payables

*Important: **cash** that is not needed for the everyday operation is **not part of the NWC**.*

2. Market value vs book value of equity

*→ difference is because book value shows the **historical costs**; valuable assets are not captured (eg. quality of management team)*

3. Market value of equity (E) = market capitalization = shares outstanding x market price per share

*→ value depends on what investors **expect in the future**.*

Balance Sheet IV.

Market to book ratio = market value of equity / book value of equity = $MV / BV = E / BV$ **OR**

price to book ratio = P / B (*in this case we measure it on a single share's level*)

→ low: value stock

→ high: growth stock

Enterprise value (EV) = Total enterprise value (TEV) → *operational value of the company*

= Market value of equity + Debt – cash (*or any Non-Operating Asset (NOA) eg. marketable securities*)

Balance Sheet VI. – example 2.2

Problem

Rylan Enterprises has 5 million shares outstanding. The market price per share is \$98. The firm's book value of equity is \$50 million.

What is Rylan's market capitalization?

What is Rylan's Market-to-Book ratio?

Balance Sheet VI. – example 2.2 – solution

Rylan's market capitalization is \$490 million

5 million shares $\times$ \$98 share = \$490 million.

$$\frac{\$490}{\$50} = 9.80.$$

Income Statement – „*bottom line*” 😊

Lists the firm's revenues and expenses over a period of time.

GLOBAL CONGLOMERATE CORPORATION

Income Statement

Year Ended December 31 (in \$ million)

	2015	2014
Total sales	186.7	176.1
Cost of sales	(153.4)	(147.3)
Gross Profit	33.3	28.8
Selling, general, and administrative expenses	(13.5)	(13.0)
Research and development	(8.2)	(7.6)
Depreciation and amortization	(1.2)	(1.1)
Operating Income	10.4	7.1
Other income	—	—
Earnings Before Interest and Taxes (EBIT)	10.4	7.1
Interest income (expense)	(7.7)	(4.6)
Pretax Income	2.7	2.5
Taxes	(0.7)	(0.6)
Net Income	2.0	1.9
Earnings per share:	\$0.556	\$0.528
Diluted earnings per share:	\$0.526	\$0.500

Bottom line...

EPS = Net income /
shares outstanding

Diluted EPS → calculates
also with employers' and
executives' **stock
options**, and
convertible bonds
(new shares would be
issued)

Statement of Cash Flows I. / a

Utilizes the information from the BS and IS to determine how much **cash** the firm has **generated**, and how the cash has been **allocated** during a set period.

Operating activity →

Investment activity →

Financing activity

Adjusts net income by all non-cash items related to operating activities and changes in net working capital

- Depreciation – add the amount of depreciation
- Accounts Receivable – deduct the increases
- Accounts Payable – add the increases
- Inventories – deduct the increases

Payment of dividends
Sale/purchase of stocks
Changes in borrowings

Capital Expenditures (CAPEX)
Buying or selling marketable securities

Statement of Cash Flows II.

GLOBAL CONGLOMERATE CORPORATION

Statement of Cash Flows
Year Ended December 31 (in \$ million)

	2015	2014
Operating activities		
Net income	2.0	1.9
Depreciation and amortization	1.2	1.1
Other non-cash items	(2.8)	(1.0)
Cash effect of changes in		
Accounts receivable	(5.3)	(0.3)
Accounts payable	4.7	(0.5)
Inventory	(1.0)	(1.0)
Cash from operating activities	(1.2)	0.2
Investment activities		
Capital expenditures	(14.0)	(4.0)
Acquisitions and other investing activity	(7.0)	(2.0)
Cash from investing activities	(21.0)	(6.0)
Financing activities		
Dividends paid	(1.0)	(1.0)
Sale (or purchase) of stock	—	—
Increase in borrowing	24.9	5.5
Cash from financing activities	23.9	4.5
Change in cash and cash equivalents	1.7	(1.3)

Statement of Cash Flows III. – example 2.2

The Impact of Depreciation on cash flow

Problem

Suppose global had an additional \$1 million depreciation expense in 2015. if Global's tax rate on pretax income is 26%, what would be the impact of this expenses on Global's earnings?

How would it impact Global's cash balance at the end of the year?

Statement of Cash Flows IV. – example 2.2 solution

Depreciation is an operation expense, so Global's operating income, EBIT, and pretax income would fall by \$1 million. This decrease in pretax income would reduce Global's tax bill by $26\% \times \$1 \text{ million} = \0.26 million . Therefore, net income would fall by $1 - 0.26 = \$0.74 \text{ million}$.

On the statement of cash flows, net income would fall by \$0.74 million, but we would add back the additional depreciation of \$1 million because it is not a cash expense. Thus cash from operating activities would rise by $-0.74 + 1 = \$0.26 \text{ million}$. Thus, global's cash balance at the end of the year would **increase by \$0.26 million, the amount of the tax savings** that resulted from the additional depreciation expense.

Statement of Stockholders' Equity

Breaks down the stockholders' equity computed on the BS into the amount that came from issuing shares versus retained earnings.

$$\begin{aligned}\text{Change in Stockholders' Equity} &= \text{Retained Earnings} \\ &\quad + \text{Net Sales of Stock} \\ &= \text{Net Income} \\ &\quad - \text{Dividends} \\ &\quad + \text{Sales of Stock} \\ &\quad - \text{Repurchase of Stock}\end{aligned}$$

Other

Other:

- Management discussion and analysis (MD&A):
 - *Providing background to the company and any significant events that occurred. Discuss the future plans, goals, new projects, etc. + liquidity issues + off-balance sheet transactions*
- Notes to the financial statements:
 - *further details, information related to the statements. Eg. Information on debts, leases, stock-based compensation plans...*

Financial Statement Analysis I.

1. Firm with itself → firm has changed over time
2. Firm to other similar firms using a common set of financial ratios
3. Type of ratios:
 - Profitability ratios → IS
 - Liquidity ratios → BS
 - Working capital ratios → IS & BS
 - Interest coverage ratios → IS
 - Leverage ratios
 - Valuation ratios
 - Operating returns

Profitability Ratios		Leverage Ratios (continued)	
Gross Margin [28%, 42%, 65%]	$\frac{\text{Gross Profit}}{\text{Sales}}$	Debt-to-Capital Ratio [18%, 38%, 56%]	$\frac{\text{Total Debt}}{\text{Total Equity} + \text{Total Debt}}$
Operating Margin [7%, 13%, 22%]	$\frac{\text{Operating Income}}{\text{Sales}}$	Debt-to-Enterprise Value Ratio [–4%, 9%, 25%]	$\frac{\text{Net Debt}}{\text{Enterprise Value}}$
EBIT Margin [6%, 12%, 20%]	$\frac{\text{EBIT}}{\text{Sales}}$	Equity Multiplier (book) [1.7x, 2.5x, 4.0x]	$\frac{\text{Total Assets}}{\text{Book Value of Equity}}$
Net Profit Margin [2%, 7%, 14%]	$\frac{\text{Net Income}}{\text{Sales}}$	Equity Multiplier (market) [1.0x, 1.1x, 1.5x]	$\frac{\text{Enterprise Value}}{\text{Market Value of Equity}}$
Liquidity Ratios		Valuation Ratios	
Current Ratio [1.2x, 1.8x, 2.9x]	$\frac{\text{Current Assets}}{\text{Current Liabilities}}$	Market-to-Book Ratio [1.6x, 2.9x, 5.5x]	$\frac{\text{Market Value of Equity}}{\text{Book Value of Equity}}$
Quick Ratio [0.7x, 1.2x, 2.0x]	$\frac{\text{Cash \& Short-term Investments} + \text{Accounts Receivable}}{\text{Current Liabilities}}$	Price-Earnings Ratio [15.7x, 21.6x, 32.6x]	$\frac{\text{Share Price}}{\text{Earnings per Share}}$
Cash Ratio [0.1x, 0.4x, 0.8x]	$\frac{\text{Cash}}{\text{Current Liabilities}}$	Enterprise Value to Sales [1.3x, 2.4x, 4.3x]	$\frac{\text{Enterprise Value}}{\text{Sales}}$
Working Capital Ratios		Enterprise Value to EBIT [11.9x, 15.7x, 22.2x]	$\frac{\text{Enterprise Value}}{\text{EBIT}}$
Accounts Receivable Days [32, 49, 67]	$\frac{\text{Accounts Receivable}}{\text{Average Daily Sales}}$	Enterprise Value to EBITDA [8.8x, 11.5x, 15.4x]	$\frac{\text{Enterprise Value}}{\text{EBITDA}}$
Accounts Payable Days [25, 42, 62]	$\frac{\text{Accounts Payable}}{\text{Average Daily Cost of Sales}}$	Operating Returns	
Inventory Days [24, 54, 92]	$\frac{\text{Inventory}}{\text{Average Daily Cost of Sales}}$	Asset Turnover [0.3x, 0.6x, 1.1x]	$\frac{\text{Sales}}{\text{Total Assets}}$
Interest Coverage Ratios		Return on Equity (ROE) [4%, 11%, 19%]	$\frac{\text{Net Income}}{\text{Book Value of Equity}}$
EBIT/Interest Coverage [2.9x, 6.7x, 15.8x]	$\frac{\text{EBIT}}{\text{Interest Expense}}$	Return on Assets (ROA) [–1%, 3%, 8%]	$\frac{\text{Net Income} + \text{Interest Expense}}{\text{Book Value of Assets}}$
EBITDA/Interest Coverage [5.2x, 9.8x, 20.2x]	$\frac{\text{EBITDA}}{\text{Interest Expense}}$	Return on Invested Capital (ROIC) [7%, 12%, 21%]	$\frac{\text{EBIT} (1 - \text{Tax Rate})}{\text{Book Value of Equity} + \text{Net Deb}}$
Leverage Ratios			
Debt-Equity Ratio (book) [21%, 60%, 121%]	$\frac{\text{Total Debt}}{\text{Book Value of Equity}}$		
Debt-Equity Ratio (market) [6%, 21%, 51%]	$\frac{\text{Total Debt}}{\text{Market Value of Equity}}$		

Financial Statement Analysis II.

Profitability ratios

Gross margin = gross profit / sales → *shows the ability to sell a product for more than the cost of production*

Operating margin = operating income / sales → *efficiency of operation*

EBIT margin = EBIT / sales → *efficiency of operation; different corporate strategy*

Net profit margin = net income / sales → *shows the fraction of each dollar in revenues that is available for equity holders (difference can come from efficiency OR leverage, OR accounting assumptions)*

Financial Statement Analysis III.

Liquidity ratios

The firm has sufficient working capital to meet its short-term needs.

Current ratio = current assets / current liabilities

Quick ratio = (cash + short term inv. + acc.receivable) / current liabilities

Cash ratio = cash / current liabilities

Financial Statement Analysis IV.

Working capital ratios

How efficiently the firm is utilizing its net working capital.

AccRec. days = $AR * 365 / \text{sales}$ → *the speed at which a company turns sales into cash*

AccPay. days = $AP * 365 / \text{cost of sales}$

Inventory days = $\text{inventory} * 365 / (\text{cost of sales} + \text{SG\&A costs})$

AR turnover = Sales / AR

AP turnover = $\text{Cost of sales} / AP$

Inventory turnover = $(\text{Cost of sales} + \text{SG\&A costs}) / \text{inventory}$

Financial Statement Analysis V.

Interest coverage ratios

$\text{EBIT} / \text{interest}$

$\text{EBITDA} / \text{interest}$

(EBITDA = EBIT + depreciation and amortization)

Financial Statement Analysis V.

Leverage ratios

The extent to which the company relies on debt as a source of financing.

Debt-Equity ratio = Total Debt / Total Equity

Debt-to-Capital ratio = Total Debt / (Total Equity + Total Debt)

Net Debt = Total Debt – Excess cash and short-term investment

Debt-to-Enterprise value = Net Debt / (Market Value of Equity + Net Debt)

Equity multiplier = Total Assets / Book value of Equity

Financial Statement Analysis VI.

Valuation ratios

Price-earnings (P/E) = $\text{Market capitalization}_{t-1} / \text{Net income}_t = \text{Share price}_{t-1} / \text{EPS}_t$!!!!

Sensitive to the choice of leverage, and depreciation and amortization.

EV / Sales

EV / EBITDA

EV/EBIT

Financial Statement Analysis VII.

Operating returns

Return on Equity (ROE) = $\text{Net Income}_t / \text{Book Value of Equity}_{t-1}$

ROE provides a measure of the return that the firm has earned on its past investments.

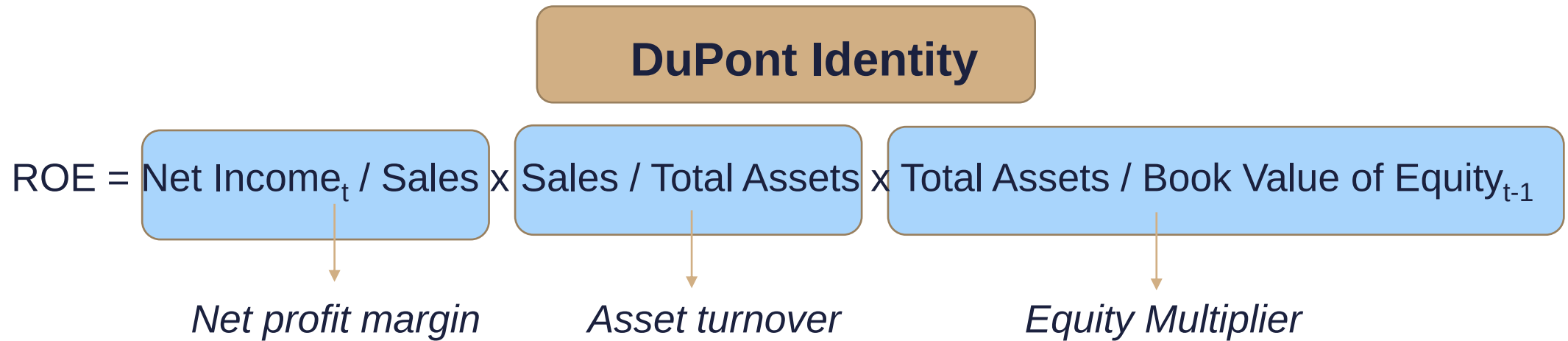
Return on Assets (ROA) = $(\text{Net Income} + \text{Interest expense})_t / \text{Book Value of Assets}_{t-1}$

Return on Invested Capital (ROIC) = $\text{EBIT} \times (1 - \text{tax rate})_t / (\text{Book Value of Equity}_{t-1} + \text{Net Debt}_{t-1})$

ROIC is the most useful in assessing the performance of the underlying business.

$\text{EBIT} \times (1 - \text{tax rate}) = \text{NOPLAT} = \text{Net Operating Profit Less Adjusted Taxes}$

Financial Statement Analysis VIII.



Financial reporting in practice

Examples for accounting scandals: Enron, WorldCom

Sarbanes-Oxley Act (SOX) in 2002 → *improve the accuracy of information given to board and shareholders*

- overhauling **incentives** and the **independence** in the **auditing** process
- stiffening **penalties** for providing **false information**
- forcing companies to **validate** their internal **financial control** processes

<https://www.congress.gov/107/plaws/publ204/PLAW-107publ204.pdf>

Dodd-Frank Act in 2010 → *response for the global financial crises of 2007/2009*

Important components eg. Financial stability, Volcker Rule (investment of banks, limiting speculative and eliminating proprietary trading)...

https://www.cftc.gov/sites/default/files/idc/groups/public/@swaps/documents/file/hr4173_enrolledbill.pdf

Exercise – Starbucks pp.85/3.

Find the most recent financial statements for Starbucks Corporation (SBUX) using the following sources:

- Company' Web page
www.starbucks.com (investor relations)
- The SEC Web site www.sec.gov (company filings in the EDGAR database)
- Yahoo! Finance Web site
finance.yahoo.com
- One other source, eg google.com (search for SBUX 10K)



View Filing Data

SEC Home » Search the Next-Generation EDGAR System » Company Search » Current Page

STARBUCKS CORP (Filer) CIK: 0000829224

Print Document View Excel Document

10-K	Cover Page - USD (\$) shares in Millions, \$ in Millions	12 Months Ended Oct. 03, 2021
Cover	Cover [Abstract]	
Cover Page	Document Type	10-K
Financial Statements	Document Annual Report	true
Notes to Financial Statements	Document Period End Date	Oct. 03, 2021
Accounting Policies	Document Transition Report	false
Notes Tables	Entity File Number	0-20322
Notes Details	Entity Registrant Name	Starbucks Corporation
All Reports	Entity Incorporation, State or Country Code	WA
	Entity Tax Identification Number	91-1325671
	Entity Address, Address Line One	2401 Utah Avenue South
	Entity Address, City or Town	Seattle
	Entity Address, State or Province	WA
	Entity Address, Postal Zip Code	98134

Starbucks Corporation (SBUX)

NasdaqGS - NasdaqGS Real Time Price. Current

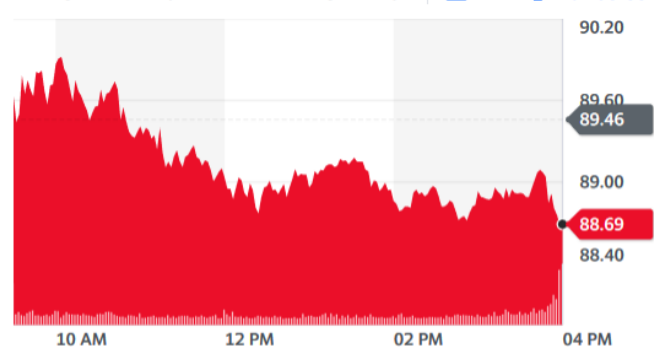
88.69 -0.77 (-0.86%)

At close: September 9 04:00PM EDT

Summary Chart Conversations Statistics Historical Data Profile Financials Analysis Options Holders Sustainability

Previous Close	89.46	Market Cap	104.781B
Open	89.50	Beta (5Y Monthly)	0.95
Bid	88.62 x 800	PE Ratio (TTM)	24.98
Ask	88.70 x 900	EPS (TTM)	3.55
Day's Range	88.64 - 89.95	Earnings Date	Nov 03, 2022
52 Week Range	68.39 - 120.15	Forward Dividend & Yield	1.96 (2.36%)
Volume	7,093,031	Ex-Dividend Date	Aug 11, 2022
Avg. Volume	6,694,170	1y Target Est	93.40

1D 5D 1M 6M YTD 1Y 5Y Max Full screen



Trade prices are not sourced from all markets



**Thank you
for your attention!**

leyla.yusifzada@uni-corvinus.hu